

How Recommendation Algorithms Decide What You See Online

We open YouTube to watch a video, but even before searching for one, our homepage is already full of videos that are relevant to our interests. However, every person has different interests. So, how are their recommendations different? For example, some people watch news and educational videos, while others watch comedy videos and movies. Their recommendations are different because behind these recommendations are algorithms that analyze information about users and the content they interact with to predict what they may want to watch.

For this system, the algorithm learns from different signals, including watch history, search history, likes, dislikes, subscriptions, “Not interested” feedback, and other information about viewer satisfaction. These signals help the system understand a user's interests and personalize the content shown to them.

Since every individual has different choices, recommendation systems use personalization and ranking. They score items and then rank them according to their predicted relevance to a particular user. Higher scores represent greater relative confidence that the user will interact with the item. As a result, different users can receive different rankings and recommendations based on their individual preferences and previous interactions.

The system can also identify a user's interests through their past viewing behaviour and information from similar viewers. If viewers have similar interests or viewing habits, the system can use this information to suggest new content that may be relevant to a particular user.

Context also matters in recommendation systems. The device a user is using and the time of day can provide additional information for recommendations. This means that recommendations can be influenced not only by what a user has interacted with in the past, but also by the context in which they interact with content.

However, recommendation algorithms do not work only through clicks. A click alone is not enough to understand whether content was actually useful or satisfying to a viewer. Clicks and viewing behaviour provide signals, while satisfaction surveys can provide additional information about viewer satisfaction. YouTube also considers long-term viewer satisfaction rather than treating a single click as the complete measure of a successful recommendation.

These technical concepts can be understood through simple examples. Imagine a librarian with thousands of books who knows your reading preferences. Instead of showing you all 1,000 books, the librarian selects and ranks the books you are most likely to enjoy. In the same way, a recommendation system does not show every available piece of content to every user. It selects and ranks content based on its predicted relevance.

Another example is a restaurant. A restaurant does not necessarily give every customer the exact same suggestion. It can consider what a customer usually orders and the situation before making a suggestion. Similarly, recommendation systems use information about a particular user and the context to make recommendations more relevant.

Conclusion

In a nutshell, recommendation algorithms are not simply guessing randomly. They learn from signals, analyze them, predict what may be relevant, rank different possibilities, and personalize recommendations according to the user's preferences and interactions. This is why two people using the same platform can see very different content. Our interactions with online content provide signals that influence the recommendations we receive.

References

1. **Google for Developers.** Recommendation systems overview.
<https://developers.google.com/machine-learning/recommendation/overview/types>
2. **YouTube Help.** How YouTube recommendations work. <https://support.google.com/youtube/answer/16089387>
3. **YouTube Help.** YouTube's recommendation system. <https://support.google.com/youtube/answer/16533387>
4. **Amazon Web Services.** How recommendation scoring works — Amazon Personalize.
<https://docs.aws.amazon.com/personalize/latest/dg/recommendations.html>
5. **Amazon Web Services.** Getting a personalized ranking — Amazon Personalize.
<https://docs.aws.amazon.com/personalize/latest/dg/rankings.html>

Fact-Check Log

Claim 1 — Recommendation systems commonly use candidate generation, scoring, and re-ranking to select and order recommendations.

Source: Google for Developers — Recommendation systems overview

Verification: Verified. Google's documentation describes candidate generation, scoring, and re-ranking as the three-stage architecture commonly used in recommendation systems.

Claim 2 — YouTube uses signals such as watch history, search history, subscriptions, likes, dislikes, “Not interested” feedback, and satisfaction surveys in its recommendation system.

Source: YouTube Help — How YouTube recommendations work

Verification: Verified. YouTube's official documentation lists these as recommendation signals and explains that satisfaction surveys help the system understand satisfaction, not only watch time.

Claim 3 — Higher recommendation scores represent greater relative confidence that a user will interact with an item.

Source: Amazon Web Services — Amazon Personalize recommendation scoring

Verification: Verified. AWS states that recommendation scores represent relative certainty about whether a user will interact with an item next, with higher scores representing greater certainty.